

PRODUCT REQUIREMENTS DOCUMENT

Sports Facility Booking

Habit Health — Engagement Module

Product	Habit Health — Engagement Module
Feature	Sports Facility Booking
Document Type	Product Requirements Document (PRD)
Status	FINAL — Ready for Development

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1. Executive Summary

1.1 Problem Statement

Employees at corporate campuses currently lack a self-service mechanism to book sports facilities. The existing process relies on manual sign-ups, phone calls, or walk-in arrangements, creating operational inefficiency for both employees and facility managers. There is no auditable trail of bookings, making kit issuance and access management error-prone.

1.2 Solution Overview

- Slot-based booking system embedded within Habit Health
- Email confirmation as the official proof of booking for facility access and sports kit collection
- Admin-controlled slot management with configurable capacity, timing, and eligibility rules
- Automated cancellation flow that releases capacity and notifies all stakeholders immediately
- CUG + User Group based eligibility ensuring only authorised employees see and book facilities
- My Bookings section for users to track all their bookings (Upcoming, Past, Cancelled)

2. Scope

2.1 Included in This Release

Feature	Description
Sports Facility Module	Top-level navigation item in Habit Health (Web + Mobile)
Slot-based Booking	Users can book available slots within configured limits
Multiple Slot Bookings	Users can book multiple slots (admin-controlled per-day and per-event limits)
User-initiated Cancellation	Cancel bookings up to the moment slot starts
Confirmation Emails	Booking and cancellation emails to user + CC to admin-configured recipients
Admin Facility Management	Create, configure, and manage sports facilities
Admin Slot Management	Create recurring slots, edit capacity, manage availability
Email CC Configuration	Admin configures CC recipients per facility
CUG + User Group Eligibility	Dual-condition access control
My Bookings Section	User view of Upcoming, Past, and Cancelled bookings
Timezone	IST (Indian Standard Time) only

2.2 Explicitly Out of Scope (Future Phases)

Feature	Notes
Admin booking console (view + cancel)	Email-based for current phase
No-show tracking	Manual tracking for current phase
Attendance marking	SPOC manual process for current phase
Reporting / data export	Analytics pipeline required
Push / in-app notifications	Notification module dependency
Waitlist / buffer bookings	Complexity deferred
Payment integration	Facilities are free
Multi-timezone support	IST only

3. Eligibility & Access Control

Sports Facility visibility is gated by a dual-condition check. Both conditions must be satisfied for a user to see any facility.

Condition 1: User belongs to the CUG (Closed User Group) assigned to the facility

AND

Condition 2: User belongs to the User Group selected for the facility

Failing either condition → Facility is completely hidden from the user

3.1 Rules Summary

Rule	Behaviour
User Group assignment is mandatory	Admin cannot publish a facility without selecting a User Group
CUG assignment is mandatory	Admin must assign at least one CUG to the facility
User in multiple User Groups	User sees facilities assigned to ANY of their User Groups
Non-eligible user visits Sports module	Empty state displayed: "No sports facilities are available..."
User loses eligibility after booking	Existing bookings preserved; can view and cancel; new bookings blocked
Admin can only see own CUGs	Admin role is scoped to their assigned CUGs

4. User-Side Product Requirements

4.1 Navigation

Sports Facility is a top-level navigation item in both Web and Mobile interfaces.

Platform	Entry Point	Landing Page
Web	Top-level menu item in main navigation	Sports Facility Listing Page
Mobile (iOS + Android)	Separate menu item in mobile navigation	Sports Facility Listing Page

4.2 Sports Facility Listing Page

Displays all eligible sports facilities for the user's CUG and User Group.

4.2.1 Facility Card Components

- Facility image (uploaded by admin)
- Status badge: Available | Fully Booked
- Title: Sport Type — Facility Name (e.g., "Badminton — Court 2")
- Campus name (display only)
- Event start date and end date

4.2.2 Hidden on Listing Card

- Slot counts or remaining capacity
- Participant count
- Search, filter, or sort controls

4.2.3 Status Logic on Listing Card

Condition	Status Shown on Card
At least one future slot has capacity	Available (green badge)
All future slots are at full capacity	Fully Booked (red badge)
Event end date has passed	Facility hidden entirely
Facility marked INACTIVE by admin	Facility hidden entirely

4.3 Facility Detail Page

Opened when the user taps a facility card. Displays full information and enables slot selection and booking.

4.3.1 Page Components

- Facility banner image(s)
- Facility name and description
- Rules / Requirements section
- Date selector (defaults to today; navigates forward only)
- Slot list for the selected date

4.3.2 Slot Card Components

- Time range (e.g., "7:00 AM — 8:00 AM")
- Remaining availability label (e.g., "3 slots left")
- Disabled state when slot is full
- "Booked" state when the current user has already booked that slot
- Disabled state when slot time has already started

4.4 Booking Flow

4.4.1 Flow Steps

1. User navigates to a facility and selects a date
2. User selects an available slot from the list
3. User taps "Book Slot" (confirmation required if multiple slots are being booked)
4. System validates: eligibility, capacity, daily/weekly limits, no overbooking, slot not started
5. On success: slot is marked Booked, success message shown, confirmation email sent
6. User can immediately book another available slot if within limits

4.4.2 Booking Rules

Rule	Behaviour
Same-day booking	Allowed
Minimum advance booking time	None — user can book any time before slot starts
Booking cut-off	When slot starts (real-time)
One booking per user per slot	Strictly enforced — user cannot book the same slot twice
Multiple slots per day	Allowed (within admin-configured daily limit)
Multiple slots per event	Allowed (within admin-configured event limit)

4.4.3 Booking Limits

Limit Type	Default Value	Configurability
Max slots per user per day	Admin configurable	Per facility via admin console
Max slots per user per event	10 slots	System default; adjustable per facility
Multi-day bookings	Allowed	No restriction
Multiple slots same day	Allowed (within daily limit)	Controlled by daily limit setting

Anti-Overbooking Guarantee: Booking is atomic. Slot capacity is the sole authority. No buffer registrations are permitted. If a slot reaches capacity between the user viewing it and submitting, the booking is rejected with a clear error message.

4.5 Cancellation Flow

4.5.1 Flow Steps

1. User locates a Booked slot (via My Bookings tab or Facility Detail page)
2. User taps "Cancel Booking"
3. System displays a confirmation prompt
4. On confirmation: slot is released, capacity increments, cancellation email is sent
5. Slot immediately becomes bookable by other users

4.5.2 Cancellation Rules

Rule	Behaviour
Cancellation deadline	Up to the last minute before slot starts
After slot starts	Cancellation NOT possible
Admin intervention	No manual admin intervention required for user-initiated cancellations
Admin notification	Admin is notified via the CC'd cancellation email

4.6 My Bookings Section

4.6.1 Location

My Bookings is a tab within the Sports Facility section (not a separate navigation item).

4.6.2 Tabs/Sections

Tab	Content	Actions Available
Upcoming	Future bookings (slot time not yet started)	Cancel
Past	Completed bookings (slot time has passed)	View only
Cancelled	User-cancelled bookings	View only

4.6.3 Booking Card in My Bookings

- Facility name
- Campus name
- Slot date
- Slot time (start — end)
- Status badge (Upcoming / Completed / Cancelled)
- Cancel button (Upcoming only)

4.6.4 Data Retention

Data Type	Retention Period
Past bookings	Forever (visible to user indefinitely)
Cancelled bookings	Forever (visible to user indefinitely)

4.6.5 Re-booking Behaviour

If a user cancels a booking and then books the same slot again: the cancelled booking remains in the "Cancelled" tab, the new booking appears in "Upcoming" (or "Past" if completed), and both records are maintained separately.

4.7 Facility Access Process

The booking confirmation email serves as the official access credential at the facility.

1. User arrives at the sports facility
2. User presents the booking confirmation email to the Clinic SPOC / Facility Manager
3. User submits their official ID card to the SPOC
4. Sports kit is issued to the user

5. ID card is returned to the user once the sports kit is handed back

The email acts as the single proof of booking. No QR code or app check-in is required for this phase.

5. Admin-Side Product Requirements

5.1 Facility Creation

Sports Facility is created as a new Event Type within the Admin Console. The Slot-based toggle is ON by default and cannot be changed.

5.1.1 Facility Configuration Fields

Field	Required	Validation
Event Name	Yes	Max 100 characters
CUG	Yes	Admin sees own CUGs only
User Group	Yes	Must select at least one
Event Start Date	Yes	Cannot be in the past
Event End Date	Yes	Must be $\geq$ start date
Registration Start Date	Yes	Must be within event dates
Registration End Date	Yes	Must be within event dates
Description / Instructions	No	Rich text recommended
Map URL	No	Valid URL format
Banner Image(s)	No	Existing multi-image system
Max Slots per User per Day	Yes	Integer ≥ 1
Max Slots per User per Event	Yes	Integer ≥ 1 , default 10
Email CC List	No	Comma-separated valid emails

5.1.2 Facility Creation Constraints

Constraint	Behaviour
Event start date in past	Blocked — admin cannot create
Registration dates outside event dates	Blocked — validation error
No User Group selected	Blocked — cannot publish
No CUG selected	Blocked — cannot publish

5.2 Slot Management

Admin can create, edit, and delete slots for each facility. Slot capacity is the sole authority for booking availability.

5.2.1 Recurring Slot Creation

Admin defines a slot rule that auto-generates individual slot instances:

Field	Description	Required
Time range	Start time — end time	Yes
Duration	Slot duration in minutes	Yes (auto-calculated)
Capacity	Maximum bookings per slot	Yes (must be ≥ 1)
Days of week	e.g., Monday, Wednesday, Friday	Yes
Effective date range	Within event start/end window	Yes

5.2.2 Slot Operations

Operation	Behaviour & Constraints
Create slots	Creates individual slot instances per recurring rule
Edit slot time/duration	Only for future slots with no bookings
Edit slot capacity	Can increase anytime; can reduce only if new capacity $\geq$ current bookings
Delete slot	NOT ALLOWED if slot has existing bookings
Overlapping slots	NOT ALLOWED — system blocks overlapping time ranges
Capacity = 0	NOT ALLOWED — minimum capacity is 1

5.2.3 Capacity Increase on Full Slot

When admin increases capacity on a "Fully Booked" slot, the slot status automatically updates to "Available" and additional capacity becomes immediately bookable.

5.3 Facility Status Management

Status	Description	User Visibility
ACTIVE	Facility is live and bookable	Visible (if eligible)
INACTIVE	Admin has disabled the facility	Hidden
EXPIRED	Event end date has passed	Hidden

5.3.1 Unpublish Constraints

Scenario	Behaviour
Facility has existing bookings	Admin CANNOT unpublish/deactivate
All bookings completed or cancelled	Admin can unpublish

5.4 Email CC Configuration

For each Sports Facility, admin can configure a list of email addresses that will be CC'd on every booking and cancellation confirmation email. Multiple email addresses can be entered as a comma-separated list. Typical recipients include Facility Managers, Clinic SPOCs, and Internal Ops Teams. CC list is per-facility and editable at any time; changes apply to future emails only.

6. Email Notification Requirements

Email is the official record and access proof for Sports Facility bookings. Emails must be reliable, well-formatted, and delivered immediately after a booking or cancellation event. All dates and times in emails are displayed in IST (Indian Standard Time).

6.1 Booking Confirmation Email

Property	Value
Trigger	Immediately after successful slot booking; one email per booking
To	User's registered email address
CC	Admin-configured email list for the facility
Subject	Sports Facility Booking Confirmed {{event_name}} {{slot_date}} {{slot_time}}

6.2 Cancellation Confirmation Email

Property	Value
Trigger	Immediately after successful slot cancellation (user-initiated)
To	User's registered email address
CC	Admin-configured email list for the facility
Subject	Sports Facility Booking Cancelled {{event_name}} {{slot_date}} {{slot_time}}

6.3 Email Variables Reference

Variable	Data Source	Example
{{user_name}}	User profile — display name	Rahul Sharma
{{user_email}}	User profile — email	rahul@company.com
{{event_name}}	Facility configuration — name	Badminton — Court 2
{{campus_name}}	Facility configuration — campus	Noida Sector-126
{{slot_date}}	Slot record — date	15 Jan 2026
{{slot_start_time}}	Slot record — start time	7:00 AM
{{slot_end_time}}	Slot record — end time	8:00 AM
{{admin_contact_emails}}	Facility config — CC email list	ops@company.com

6.4 Email Non-Functional Requirements

- Emails must not block the booking/cancellation flow (async delivery)
- Delivery failure must be logged and retried; must not result in booking failure
- All emails must be auditable (sent timestamp, recipient list, template version)
- CC list must be dynamically resolved at send time (not cached at booking time)
- All times displayed in IST

7. Status & Visibility Rules

Condition	Status / Behaviour	Where Displayed
At least one future slot has capacity	Available (green badge)	Listing card
All future slots at full capacity	Fully Booked (red badge)	Listing card
Event end date has passed	Facility hidden (EXPIRED)	Not shown
Admin marks facility INACTIVE	Facility hidden	Not shown
All slots on selected date are full	"No slots available" message	Facility detail page
User not in CUG or User Group	Facility hidden; empty state	Not shown
Registration window not yet open	Slot selection disabled	Facility detail page
Registration window closed	Booking unavailable	Facility detail page
Slot time has already started	Slot disabled for booking	Facility detail page
User has already booked a slot	"Booked" badge on slot	Facility detail page

8. Non-Functional Requirements

8.1 Performance

Requirement	Target	Notes
Listing page load time	<2 seconds (P95)	Cached facility data acceptable
Slot list load time	<1.5 seconds (P95)	Real-time capacity needed
Booking confirmation time	<3 seconds end-to-end	Includes DB write + email trigger
Concurrent booking safety	0 overbooking under load	Atomic DB operations required

8.2 Reliability & Data Integrity

- Booking must be atomic: capacity decrement and booking record creation in a single operation
- No overbooking is permitted under any concurrency scenario
- Slot capacity reduction below current bookings must be blocked at the API level
- Audit log must capture all booking and cancellation events with timestamps and actor IDs
- All mutations must include before/after state for reconciliation

8.3 Security & Access Control

- API endpoints must enforce RBAC (Role-Based Access Control) via @PreAuthorize annotations
- Users can only view, book, and cancel their own bookings
- Admins can only manage facilities within their assigned CUGs
- CUG + User Group eligibility must be enforced server-side, not client-side only
- Email CC list must not expose PII to unauthorised parties

8.4 Timezone

IST (Indian Standard Time) only. No timezone conversion or multi-timezone support. All dates and times stored and displayed in IST.

9. Data Model Requirements

The following describes the required data entities and key fields. Detailed MongoDB schema design is the responsibility of the backend team.

9.1 Sports Facility Entity

Field	Type	Required	Notes
id	ObjectId	Yes	Primary key
cug_code	String	Yes	CUG assignment; scopes visibility
user_group_ids	Array<String>	Yes	User Group eligibility
name	String	Yes	Display name of facility
description	String	No	Rich text; rules and instructions
campus_name	String	Yes	Display-only campus label
event_start_date	Date	Yes	Visibility window start
event_end_date	Date	Yes	Visibility window end
max_slots_per_user_per_day	Integer	Yes	Admin configurable; minimum 1
max_slots_per_user_per_event	Integer	Yes	Default 10; minimum 1
email_cc_list	Array<String>	No	CC recipients for emails
status	Enum	Yes	ACTIVE / INACTIVE / EXPIRED

9.2 Slot Entity

Field	Type	Required	Notes
id	ObjectId	Yes	Primary key
facility_id	ObjectId	Yes	FK to Sports Facility
slot_date	Date	Yes	Date of this slot instance
start_time	Time	Yes	Slot start time (IST)
end_time	Time	Yes	Slot end time (IST)
capacity	Integer	Yes	Total bookable seats; minimum 1
booked_count	Integer	Yes	Current bookings; must be atomic
status	Enum	Yes	AVAILABLE / FULL / CANCELLED

9.3 Booking Entity

Field	Type	Required	Notes
id	ObjectId	Yes	Primary key

facility_id	ObjectId	Yes	FK to Sports Facility
slot_id	ObjectId	Yes	FK to Slot
user_id	ObjectId	Yes	FK to User
user_email	String	Yes	Denormalised for email sending
user_name	String	Yes	Denormalised for email sending
status	Enum	Yes	CONFIRMED / CANCELLED
booked_at	DateTime	Yes	Booking creation timestamp
cancelled_at	DateTime	No	Null if not cancelled

10. API Requirements

All endpoints must enforce RBAC and CUG scoping.

10.1 User-Facing APIs

Method	Endpoint	Purpose
GET	/api/sports-facilities	List eligible facilities for user's CUG + User Group
GET	/api/sports-facilities/:id	Get facility details including description and rules
GET	/api/sports-facilities/:id/slots?date={date}	Get all slots for a facility on a given date
POST	/api/sports-facilities/:id/slots/:slotId/book	Book a slot; atomic capacity decrement
DELETE	/api/sports-facilities/bookings/:bookingId	Cancel own booking; release capacity
GET	/api/sports-facilities/my-bookings	List user's bookings (all statuses)
GET	/api/sports-facilities/my-bookings?status={status}	Filter by status: upcoming, past, cancelled

10.2 Admin-Facing APIs

Method	Endpoint	Purpose
POST	/api/admin/sports-facilities	Create a new Sports Facility
PUT	/api/admin/sports-facilities/:id	Update facility configuration
PATCH	/api/admin/sports-facilities/:id/status	Update facility status (ACTIVE/INACTIVE)
DELETE	/api/admin/sports-facilities/:id	Archive/deactivate facility (only if no bookings)
POST	/api/admin/sports-facilities/:id/slots	Create slots (single or recurring rule)
PUT	/api/admin/sports-facilities/:id/slots/:slotId	Edit slot capacity, time, or duration
DELETE	/api/admin/sports-facilities/:id/slots/:slotId	Delete slot (only if no bookings)
PUT	/api/admin/sports-facilities/:id/email-cc	Update CC email list for a facility

Security critical: All admin endpoints must include @PreAuthorize annotations enforcing role and CUG scope.

11. Edge Cases & Error Handling

Edge Case	Expected Behaviour	User Message
Slot fills between view and booking	Booking rejected	This slot is no longer available.

User tries to book same slot twice	Booking rejected	You have already booked this slot.
User exceeds daily booking limit	Booking rejected	You have reached the maximum bookings for today.
User exceeds event booking limit	Booking rejected	You have reached the maximum bookings for this facility.
User tries to book after slot starts	Booking rejected	This slot has already started.
User tries to cancel after slot starts	Cancellation rejected	Cannot cancel a slot that has already started.
Registration window not open	Slot selection disabled	Bookings open on [date].
Registration window closed	Slot selection disabled	Booking period for this facility has ended.
Admin tries to delete slot with bookings	Operation blocked	Cannot delete slot with existing bookings.
Admin tries to reduce capacity below bookings	Operation blocked	Cannot reduce capacity below current booking count.
Admin tries to create overlapping slots	Operation blocked	Slot times cannot overlap with existing slots.
Admin tries to create facility with past start	Operation blocked	Event start date cannot be in the past.
Admin tries to create slot with capacity = 0	Operation blocked	Slot capacity must be at least 1.
Admin tries to unpublish with existing bookings	Operation blocked	Cannot unpublish facility with existing bookings.
Email delivery fails	Booking still confirmed; email requeued	User-facing message; logged for ops

12. Acceptance Criteria

12.1 User-Side

- Eligible users can see the Sports Facility module in the main navigation on Web and Mobile
- Non-eligible users see the empty state; facility is not accessible via direct URL
- Facility listing page renders all eligible facilities with correct status badges
- Facility detail page shows rules, date selector, and real-time slot availability
- User can successfully book a slot and receives a confirmation email within 60 seconds
- User cannot book beyond their daily or event slot limit
- User cannot book the same slot twice
- User cannot book a slot after it has started
- User can cancel a booking up to the last minute before slot starts
- User cannot cancel after slot starts
- User receives a cancellation email within 60 seconds
- No overbooking occurs under concurrent booking load
- My Bookings section displays Upcoming, Past, and Cancelled tabs
- User can cancel from My Bookings section
- Past and cancelled bookings are retained forever

12.2 Admin-Side

- Admin can create a Sports Facility with all required fields via the Admin Console
- Admin cannot create a facility with event start date in the past
- Admin cannot publish a facility without CUG and User Group
- Admin can create recurring slots and view generated slot instances
- Admin cannot create overlapping slots
- Admin cannot create slots with capacity = 0
- Admin cannot delete a slot with existing bookings
- Admin cannot reduce slot capacity below current booking count
- Admin cannot unpublish a facility with existing bookings
- Admin can configure a CC email list; all future emails include the CC list
- Admin cannot set registration dates outside event dates

12.3 Email

- Booking confirmation email is sent to user with all correct template variables populated
- Admin-configured CC list receives a copy of all booking and cancellation emails
- Cancellation confirmation email is sent immediately after cancellation
- Email delivery failure does not cause booking to fail
- All times in emails are displayed in IST

12.4 Security

- All API endpoints enforce RBAC via @PreAuthorize annotations
- CUG and User Group eligibility is enforced server-side
- Users can only cancel their own bookings
- Admins can only manage facilities within their assigned CUGs

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